SET-1:

1. Write the python codes for the following situations.
   a. A biologist models a species where each month's population equals the sum of the previous two months. Print the population for the first n months.
   b. A pricing engine needs a multiplier table up to 10 for a given rate. Read the rate and print its table
   c. A broker compares the monthly rent of three properties. Display which property earns the most.

2. Write the python codes for the following situations.
   a. An analyst receives a raw transaction CSV. Load it, view the first rows, check the shape, data types and count missing values. Fill the missing Age with the column mean and the missing Amount with the column median so no analysis breaks.
   b. Scale the Amount column to the range 0 to 1 (min-max) and standardize Age (z-score) before modeling. Remove transactions whose Amount lies outside 1.5 times the interquartile range (IQR) because they distort totals. Plot a bar chart of total spending per category and a line chart of monthly spending so executives can see trends and category shares.

3. Write the python codes for the following situations.
   a. An HVAC model combines four 2 x 2 zone matrices into one 4 x 4 composite matrix.
   b. A flat array of 12 sensor values must be reshaped into a 3 x 4 matrix and each element cubed.
   c. A finance team has a departmental expense matrix. Compute the total of every row and every column.

SET-2:

1. Write the python codes for the following situations.
   a. A data scientist wants to standardize a dataset of heights (mean 170 cm, std 8 cm). Generate standard normal variables, convert the height data to z-scores and verify that the z-scores have mean about 0 and standard deviation about 1.
   b. A help desk sees customers arrive with an average gap of 2 minutes (exponential waiting time). Generate the waiting times and verify that the simulated mean is close to 2 minutes.
   c. A quality engineer draws samples from a t distribution with 10 degrees of freedom. Compute the probability that a value exceeds 2 and compare it with the standard normal distribution to show the heavier tails of t.

2. Write the python program for the below scenarios.
   a. A banking backend must verify that a security PIN is prime. Write is prime(n) and test the submitted PIN.
   b. A warehouse marks a tracking number as "perfect" if the sum of its proper divisors equals the number. Check the given number.
   c. A contest portal wants the runner-up score from a list of candidate evaluations.

3. Write the python program for the following.
   a. Three sensor rows are recorded. Find the rank of the matrix and state whether the rows are linearly dependent.
   b. Two operations, a rotation-like matrix A and a scaling matrix B, are applied in different orders. Show that A x B is not equal to B x A.
   c. A multi-zone structure is modeled by two independent 2 x 2 blocks placed on the diagonal of one big matrix. Build it and verify that its determinant equals the product of the block determinants.

SET-3:

1. Write the python codes for the following situations.
   a. A message log must isolate any word that contains a digit.
   b. A word stylist converts every vowel in a document to uppercase and every consonant to lowercase.
   c. A tech team forms an acronym for each multi-word project title by taking the first and last letter of every word.

2. Write the python program for the following.
   a. From a 4 x 4 satellite image matrix, extract the central 2 x 2 region and the last column
   b. Two design matrices A (2x3) and B (3x2) must be multiplied to find the combined loading forces.
   c. A stochastic model multiplies two random 2 x 2 integer matrices (seed 7) to test noise behavior.

3. Write the python program for the below scenarios.
   a. An essay checker counts the long words (more than 6 characters) in a paragraph.
   b. An audit script counts how many identification tokens contain at least one repeated digit.
   c. An architecture tool lists all right-angle triples (a, b, c) with a < b < c and c up to a given limit.

SET-4:

1. Write the python program for the following.
   a. A finance tool lists Fibonacci milestones that stay below a threshold n. Print all of them.
   b. An encryption checker must list every Armstrong number between a lower and an upper limit.
   c. Two mailing lists must be combined into one list with no repeated address while keeping the first appearance order.

2. Write the python program for the below scenarios.
   a. A simulator generates a random 3 x 3 matrix of integers, then prints its total sum, row sums and squared elements.
   b. A routing security check needs a 4 x 4 block matrix with identity in the top-left, zeros in the top-right, zeros in the bottom-left and 2 times identity in the bottom-right.
   c. A materials lab has a stress grid. Compute the element-wise square and cube of the matrix.

3. Write the python code
   a. A store receives on average 4 customers per hour (Poisson, lambda = 4). Estimate the probability that more than 6 customers arrive in an hour by simulation and compare it with the exact value.
   b. A game club tosses a fair coin 10 times in each experiment and repeats it 1000 times. Simulate this and find the most frequent number of heads.
   c. A marketing team is uncertain about a conversion rate and models it with a Beta distribution with alpha = 2 and beta = 5. Generate samples and compute the expected probability.

SET-5:

1. A computer vision lab is processing small grayscale image patches represented as 2D numerical matrices. Develop Python programs using NumPy for the following requirements:

(a) Extract a specific sub-region from a given 4×4 image matrix by selecting rows 1 to 2 and columns 1 to 2.

(b) Reshape a flat 1D pixel stream of 9 elements into a 3×3 image patch, and then amplify its pixel intensities by computing the cube of all elements.

(c) Build a composite 4×4 test pattern by creating a block matrix consisting of a 2×2 Identity matrix, a 2×2 Zero matrix, and two constant 2×2 matrices.

2. A smart monitoring system records continuous temperature readings and receives encrypted device tokens over a network. Develop Python programs for the following requirements:

(a) A sequence of temperature readings is recorded during a test run. Write a program to find the length of the longest continuous period where temperature readings strictly increase from one observation to the next.

(b) A transmission sends two device identification codes. Determine whether one code is a circular rotation of the other code.

3. A robotics laboratory tracks autonomous robot sensor diagnostics and manages monthly spare-part billing across four robotic workgroups. Develop suitable Python programs using NumPy and generator expressions for the following requirements:

(a) A robot logs 1000 continuous sensor readings generated via a lazy generator expression. Standardize the data using its sample mean and standard deviation, and verify that the normalized dataset has zero mean and unit standard deviation.

(b) The facility stocks three types of spare parts for four robot workgroups. Given a 4×3 base cost matrix S, a 1D discount vector d of length 3, and a 2D tax vector t of shape 4×1, use NumPy broadcasting to compute the final billed amounts and determine which workgroup generated the maximum total revenue.

SET-6:

1. An examination department processes a list of raw student assessment scores to compile rank reports. Develop suitable Python programs for the following requirements:

(a) A test score list contains multiple identical marks. Eliminate all duplicate entries and arrange the unique scores from the lowest to the highest.

(b) To mitigate grading anomalies, eliminate the lowest score and the highest score from a list of marks, and then display the remaining values in reverse order of entry.

(c) Create a separate duplicate of the original marks list and sort it in descending order to publish a merit list while ensuring the original list remains unaltered.

2. A quality inspection team on a manufacturing floor logs test metrics for manufactured parts to verify variations across a production batch. Develop suitable Python programs using NumPy and Matplotlib for the following requirements:

(a) Generate 100 random integer measurement readings ranging between 1 and 100 to represent component stress values. Draw a simple line plot connecting all 100 measurements against their sample indices (1 to 100).

(b) Using the same 100 generated measurement readings, create a scatter plot displaying individual data points to highlight the dispersion of the metrics.

(c) Represent the identical 100 random readings using both a vertical bar chart and a horizontal bar chart with suitable axis titles and legends.

3. An online retail platform records monthly sales units for various electronic brands to evaluate catalog performance. Develop suitable Python programs for the following requirements:

(a) The sales units for four smartphone brands are stored in a dictionary. Find and display the total sales units across all brands.

{b) From the brand sales record, filter and display only those brands that have sold more than 50 units using dictionary comprehension.

(c) Identify and display the specific brand that recorded the maximum sales volume during the month.

SET-7:

1. An examination committee is processing student test results to generate automated performance reports. Develop suitable Python programs for the following requirements:

a) Accept a list of marks obtained by a group of students and compute the average score of the batch.

(b) Define a user-defined function that accepts the marks list and returns the highest score obtained among all students.

(c) Inspect the list of student marks and determine how many students received an even numerical mark.

2. A stochastic computing laboratory runs Monte-Carlo approximations and evaluates streaming sensor data buffers. Develop suitable Python programs using NumPy and generator expressions for the following requirements:

(a) Formulate a generator of 10,000 coordinate pairs in the unit square $[0,1] \times [0,1]$ and compute the Monte-Carlo estimate of the mathematical constant π based on the points falling inside the quarter circle.

(b) Synthesize a continuous stream of 200 observation readings. Generate sliding temporal windows of length 10 using a generator expression and compute the sample covariance for each successive window.

(c) Generate 500 random sensor values lazily using np.fromiter and compute the statistical mean, variance, and median of the dataset.

3. A document processing pipeline cleans text entries and generates shorthand code representations for sensitive files. Develop suitable Python programs using list comprehensions for the following requirements:

(a) An analyst provides a list of mixed tokens containing punctuation and alphanumeric codes. Extract only the purely alphabetic words and convert them to lowercase using a list comprehension.

(b) In an input sentence, locate all words having a length strictly greater than 4 characters and reverse each of them while leaving other words out.

(c) Given a list of descriptive tags, create two-letter abbreviations for every word having at least 2 characters by combining its first and last characters.

SET-8:

1. A university placement cell compiles hiring statistics across multiple academic branches and salary offerings to observe historical trends. Develop suitable Python programs using Matplotlib for the following requirements:

(a) Five academic years (1 to 5) are monitored. In Series 1, hiring counts are recorded as [40, 48, 55, 62, 70], and in Series 2, they are recorded as [35, 42, 45, 50, 58]. Draw a multiple line plot displaying both series on the same chart with circular and square markers and a clear legend

(b) A student budget breakdown allocates funds as Tuition (45%), Rent (25%), Food (18%), and Books (12%). Construct a pie chart highlighting the percentage share of each category.

(c) Five departments (A, B, C, D, E) record company visits of 18, 32, 24, 28, and 15 respectively. Construct a bar chart showing these values with proper labels and an informative title.

2. A survey company collects voting response IDs and numerical factor scores from public polls for statistical evaluation. Write Python programs for the following requirements:

(a) Given a list of n voter choice IDs, identify all distinct IDs that appear strictly more than $n/3$ times in the list.

(b) From a collection of positive and negative numerical factor scores, find the maximum possible product that can be obtained by choosing any three values.

3. A mechanical vibration laboratory analyzes natural frequencies and structural resonance modes of oscillating beams. Develop suitable Python programs using the NumPy linalg package for the following requirements:

(a) Given a symmetric stiffness matrix A, compute its eigenvalues and eigenvectors. Verify that the eigenvectors corresponding to distinct modes are mutually orthogonal by evaluating their dot product.

(b) For a given vibration transition matrix, obtain the eigenvalues and eigenvectors, and numerically demonstrate that $Av = \lambda v$ using np.allclose.

(c) Check whether a coordinate transformation matrix is singular by computing its determinant and verifying whether its inverse can be formed.